

Coarsening Confers Robustness: Causal Bayesian Optimization under DAG Misspecification

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Abstract

Causal Bayesian optimization (CBO) uses a causal graph to restrict candidate interventions and to construct priors, but assumes a correct DAG. We propose coarsening as an input contract: the practitioner supplies a partition of the manipulable variables and its quotient graph. If every graph-dependent component factors through this pair, fine-graph hypotheses with the same quotient produce identical trajectories. We instantiate the contract for three families of the field: QCBO, QDCBO, and QM-CBO. Each method is invariant to quotient-invisible edge errors and recovers its base method at the identity partition. We characterize the action-space cost of restricting interventions to unions of clusters: under full intervention support and cluster-contained confounding, coarsening preserves the best attainable value, while intervention constraints or confounding outside a cluster can create a positive action-space gap. A fixed partition with a positive gap incurs linear cumulative regret. Hierarchical refinement restores vanishing average regret once an optimal arm is exposed.

1 Introduction

Causal Bayesian optimization searches over interventions $\text{do}(\mathbf{X}_s = \mathbf{x}_s)$ and uses observational data with a causal graph to construct priors over the interventional response [Aglietti et al., 2020]. CBO and its descendants [Aglietti et al., 2021, 2023, Sussex et al., 2023] assume a correct DAG. Stress tests show that

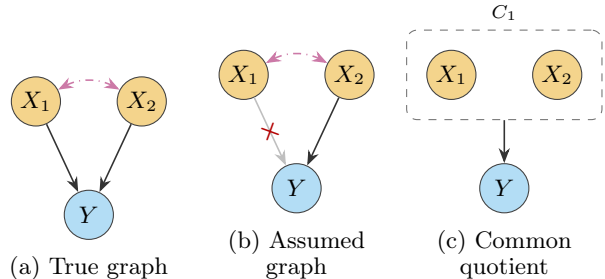


Figure 1: PARALLEL PARENT (orange: manipulable, blue: target, dash-dotted: latent confounding). Deleting $X_1 \rightarrow Y$ changes the fine search space but not the quotient: $C_1 \rightarrow Y$ survives through $X_2 \rightarrow Y$, so a quotient method receives the same input under (a) and (b).

method performance can change when only the assumed graph is perturbed [Anonymous, 2026]. We study *graph misspecification*, where the learner trusts one incorrect graph, and replace fine structure with a coarser structural commitment.

Wrong edges are costly: two-variable example. Take manipulable X_1, X_2 , latently confounded, both parents of

$$Y = \lambda(X_1 - 2)^2 + (X_2 + 2)^2 + \varepsilon, \quad y^* = 0.$$

Deleting one edge from the assumed graph removes every X_1 -containing candidate from the fine search space, and no amount of interventional data restores a candidate the search space does not contain:

$$X_1 \rightarrow Y \text{ deleted} \implies \min_{x_2} \mathbb{E}[Y \mid \text{do}(x_2)] = 4.25\lambda.$$

Commit instead to the cluster $C_1 = \{X_1, X_2\}$: the quotient remains $C_1 \rightarrow Y$ under both hypotheses (the edge survives through $X_2 \rightarrow Y$), so a quotient method’s input, and hence its trajectory, is unchanged (Fig. 1). Quotient-visible edits are not protected, and whole-cluster actions can be too coarse. §3 quantifies both boundaries.

Coarser commitments decide less, so less can be wrong. A cluster-level structural belief encodes strictly fewer decisions than a full DAG: one aggregate judgement between two groups of variables stands in for several edge-level judgements the modeler is never asked to make. Both ends of this spectrum have been studied: full-DAG CBO at one end, methods that learn structure online at heavy per-trial cost at the other [Mukherjee et al., 2024, Branchini et al., 2023]. Reliable cluster-level structure has a developed identification theory [Anand et al., 2023, Ferreira and Asaad, 2025, Wahl et al., 2024] but, to our knowledge, no optimization method. QCBO fills this gap. The practitioner supplies a partition Π of the manipulable variables and the C-DAG $\mathcal{G}^\Pi$ over its clusters (when a fine-grained hypothesis graph is available, $\mathcal{G}^\Pi$ is its quotient, with bidirected edges absorbing latent confounding; Lee and Bareinboim, 2019). QCBO explores cluster-level interventions, the minimal intervention sets of the C-DAG, using identified cluster-level do-calculus estimates as GP prior means where the quotient identifies them, and a common uninformative prior where it does not. This input never looks inside a cluster, so an edge error confined to a cluster cannot even be represented.

Contributions.

- **Robustness as an input contract.** Quotient invariance is *method-agnostic* (Theorem 1): any optimizer whose graph-dependent components factor through $(\Pi, \mathcal{G}^\Pi)$ is invariant to arbitrary within-cluster edge errors. We instantiate the contract for the arm-based, dynamic, and model-based families (QCBO, QDCBO, QMCBO; §4).
- **Action-space cost.** Coarsening restricts interventions to unions of clusters. We characterize the resulting gap $\Delta(\Pi)$, give sufficient conditions for $\Delta(\Pi) = 0$, and show a fixed partition with $\Delta(\Pi) > 0$ incurs linear cumulative regret that hierarchical refinement removes (§3).
- **Experiments.** Three minimal SCMs with closed-form oracles isolate the failure channels one at a time, followed by a portability suite running each quotient method against its base on the base paper’s own experiments (§5).

2 Background

A structural causal model $\mathcal{M} = \langle \mathbf{U}, \mathbf{V}, F, P(\mathbf{U}) \rangle$ induces a distribution over observables $\mathbf{V}$ and, for any $\mathbf{X}_s \subseteq \mathbf{V}$, interventional distributions $P(\cdot \mid \text{do}(\mathbf{X}_s = \mathbf{x}_s))$ obtained by replacing the mechanisms of $\mathbf{X}_s$ with constants [Pearl, 2009]. The induced graph $\mathcal{G}$ has a directed edge per functional argument and a bidirected

edge per shared latent. Projecting out latents yields an ADMG over $\mathbf{V}$ [Lee and Bareinboim, 2019].

Causal Bayesian optimization. Given manipulable variables $\mathbf{X} \subseteq \mathbf{V}$ and target Y , CBO [Aglietti et al., 2020] solves

$$(\mathbf{X}_s^*, \mathbf{x}_s^*) = \arg \min_{\substack{\mathbf{X}_s \in \mathbf{ES} \\ \mathbf{x}_s \in D(\mathbf{X}_s)}} \mathbb{E}[Y \mid \text{do}(\mathbf{X}_s = \mathbf{x}_s)], \quad (1)$$

where $D(\mathbf{X}_s)$ is the interventional domain, maintaining one GP per candidate set $\mathbf{X}_s$ in an exploration set $\mathbf{ES}$ (each a continuous BO sub-problem over $\mathbf{x}_s$), with prior mean the do-calculus estimate from observational data and the assumed graph. The exploration set is pruned structurally: CBO searches over the *minimal intervention sets* (MIS) of the assumed graph [Lee and Bareinboim, 2018, Aglietti et al., 2020]. Writing $\mathcal{G}_{\bar{\mathbf{X}}_s}$ for the mutilated graph with all edges into $\mathbf{X}_s$ removed and $\text{An}(\cdot)$ for ancestors,

$$\mathbf{ES} = \text{MIS}(\mathcal{G}) = \{ \emptyset \neq \mathbf{X}_s \subseteq \mathbf{X} : \mathbf{X}_s \subseteq \text{An}_{\mathcal{G}_{\bar{\mathbf{X}}_s}}(Y) \} : \quad (2)$$

if some member of $\mathbf{X}_s$ is no ancestor of Y once the whole set is intervened on, $\text{do}(\mathbf{X}_s)$ induces the same distribution as intervening on a strict subset, so the arm is redundant. The possibly-optimal sets POMIS are a further graph-only pruning with $\text{POMIS} \subseteq \text{MIS}$ [Lee and Bareinboim, 2018]. Each trial selects the arm and value maximizing cost-normalized causal expected improvement [EI; Garnett, 2023]; the model-based family MCBO selects instead by the Gaussian-process upper confidence bound acquisition function [GP-UCB; Srinivas et al., 2010, Sussex et al., 2023]. Both the pruning and the priors are functions of the assumed graph, which is where misspecification enters.

Coarsening and C-DAGs.

Definition 1 (Coarsening [Madaleno et al., 2026a]). A partition $\Pi = \{C_1, \dots, C_K\}$ of $\mathbf{V}$ induces the map $\chi : \mathbf{V} \rightarrow \Pi$ and the quotient graph $\mathcal{G}^\Pi = (\Pi, E^\Pi)$ of a graph $\mathcal{G} = (\mathbf{V}, E)$,

$$E^\Pi = \{(\chi(u), \chi(v)) : (u, v) \in E, \chi(u) \neq \chi(v)\}.$$

Π is valid when the directed part of $\mathcal{G}^\Pi$ is acyclic. $\mathcal{G}^\Pi$ is then a *coarsening* of $\mathcal{G}$.

For ADMGs the quotient additionally carries $C_k \leftrightarrow C_\ell$ iff some $V_i \in C_k, V_j \in C_\ell$ share a bidirected edge. The result is a C-DAG in the sense of Anand et al. [2023]: do-calculus on it is sound and complete for the class of fine graphs that quotient to it. Markov properties transfer to the cluster level, faithfulness does not [Wahl et al., 2024].

Four objects. Write $\Pi_M \subseteq \Pi$ for the manipulable clusters and $\mu^*(\mathbf{X}_s) = \min_{\mathbf{x}_s \in D(\mathbf{X}_s)} \mathbb{E}[Y \mid \text{do}(\mathbf{X}_s = \mathbf{x}_s)]$. The paper is organized around

$$\begin{aligned} U_\Pi(\mathcal{S}) &:= \bigcup_{C \in \mathcal{S}} C, & Q_\Pi(\mathcal{G}) &:= (\Pi, \mathcal{G}^\Pi), \\ \mathcal{U}(\Pi) &:= \{U_\Pi(\mathcal{S}) : \emptyset \neq \mathcal{S} \subseteq \Pi_M\}, \\ V(\Pi) &:= \min_{A \in \mathcal{U}(\Pi)} \mu^*(A), & \Delta(\Pi) &:= V(\Pi) - V(\Pi_\perp), \end{aligned}$$

where $\Pi_\perp$ is the all-singleton partition: the quotient input, the cluster-union interventions, the best value attainable from them, and the gap coarsening opens. Since $\mathcal{U}(\Pi) \subseteq \mathcal{U}(\Pi_\perp)$, $\Delta(\Pi) \geq 0$ (Prop. 2).

Evaluation metrics. We report the best value found y_T^{best} and the *empirical* incumbent regret $\hat{r}_t = |y_t^{\text{best}} - y^*|$ with $\hat{R}_T = \sum_{t=1}^T \hat{r}_t$, computed from observed incumbents (recorded with noise σ_Y), an empirical proxy for the incumbent mean regret of §3.2. Behavioral robustness is scored by *paired-by-seed* differences between a misspecified and a correct run of the same method (§5).

3 What coarsening buys

An optimizer meeting the contract accepts $Q_\Pi(\mathcal{G}) = (\Pi, \mathcal{G}^\Pi)$ in place of a fine graph: every graph-dependent component (search space, identification formulas, priors, surrogate architecture, selection rules) must be a function of that pair. The guarantee is input equivalence. It protects behavior against graph errors the quotient cannot see. Solution quality rests on the quotient itself being correct. Throughout, Π, Π' are manipulable-only partitions, and $\Pi' \preceq \Pi$ (Π' refines Π) iff every cluster of Π' is contained in a cluster of Π [Madaleno et al., 2026a].

3.1 What is protected?

Theorem 1 (Quotient invariance is method-agnostic). *Let $\mathcal{A}$ be a causal Bayesian optimizer that factors through the quotient: for some $\bar{\mathcal{A}}$ and every admissible hypothesis $\mathcal{G}$ (acyclic, with valid quotient under Π),*

$$\mathcal{A}(\mathcal{G}, \mathcal{D}, \omega, \eta) = \bar{\mathcal{A}}(Q_\Pi(\mathcal{G}), \mathcal{D}, \omega, \eta)$$

for all data $\mathcal{D}$, seeds ω , and hyperparameters η . Then for admissible $\mathcal{G}, \mathcal{G}'$,

$$Q_\Pi(\mathcal{G}) = Q_\Pi(\mathcal{G}') \implies \tau_{\mathcal{A}}(\mathcal{G}) = \tau_{\mathcal{A}}(\mathcal{G}'),$$

where $\tau_{\mathcal{A}}(\mathcal{G})$ is the full trajectory under hypothesis $\mathcal{G}$. (Proof: App. A.)

Within-cluster edits (insertion, deletion, or reversal of directed or bidirected edges) and inter-cluster edits redundant in the quotient satisfy the premise. A

quotient-visible edge is not protected (§5.1). The theorem covers both routes by which a wrong graph reaches an optimizer,

$$\mathcal{G} \mapsto (\mathcal{E}(\mathcal{G}), \{m_s^{\mathcal{G}}\}_{s \in \mathcal{E}(\mathcal{G})}) \mapsto \tau,$$

where $\mathcal{E}$ is the exploration set and m_s the prior. A corrupted prior is overridden by interventional data while the arm survives. A removed arm is never played again. Neither map references a quotient-invisible edge.

3.2 What does protection cost?

Proposition 1 (Validity is not refinement-monotone). *There exist a valid Π and a refinement $\Pi' \preceq \Pi$ with $\mathcal{G}^\Pi$ acyclic and $\mathcal{G}^{\Pi'}$ cyclic. Validity must be re-verified after each refinement (it remains decidable from cluster-level structure alone).*

Lemma 1 (Refinement: values persist, membership need not). *Let $\Pi' \preceq \Pi$, both valid. Every $\mathbf{X}_s \in \mathbf{ES}^\Pi$ that is again a MIS of $\mathcal{G}^{\Pi'}$ lies in $\mathbf{ES}^{\Pi'}$ with the same identification functional. Membership itself is not refinement-monotone: a coarse cluster-union arm need not be minimal at the finer partition (App. A).*

Proposition 2 (Price of coarsening). *$V(\Pi)$ is attained on the exploration set QCBO actually searches (Eq. 3, Lemma 2):*

$$\min_{A \in \mathbf{ES}^\Pi} \mu^*(A) = \min_{A \in \mathcal{U}(\Pi)} \mu^*(A) = V(\Pi).$$

For any valid $\Pi' \preceq \Pi$,

$$\mathcal{U}(\Pi) \subseteq \mathcal{U}(\Pi'), \quad 0 \leq \Delta(\Pi') \leq \Delta(\Pi),$$

and $\Delta(\Pi) = 0$ iff some optimizer of Eq. (1) lies in $\mathcal{U}(\Pi)$.

Proposition 3 (Sufficient conditions for zero coarsening gap). *Assume a semi-Markovian SCM. For a fine intervention $A \subseteq \mathbf{X}$ with value $a \in D(A)$, let*

$$\text{cl}_\Pi(A) := \bigcup_{\substack{C \in \Pi \\ C \cap A \neq \emptyset}} C, \quad B := \text{cl}_\Pi(A) \setminus A,$$

and let B_a be the joint natural response of B under $\text{do}(A = a)$. If, for every such A and a ,

- (i) $\text{supp}(B_a) \subseteq D_B(a) := \{b : (a, b) \in D(\text{cl}_\Pi(A))\}$,
- (ii) $B_a \perp Y_{a,b}$ for all $b \in D_B(a)$,

then $V(\Pi) = V(\Pi_\perp)$ and $\Delta(\Pi) = 0$. (Proof: App. A. The mixture step is $\mathbb{E}[Y_a] = \int \mathbb{E}[Y_{a,b}] dP(B_a = b) \geq \inf_{b \in D_B(a)} \mathbb{E}[Y_{a,b}]$.)

Corollary 1 (Graph-level conditions). *In a semi-Markovian SCM, if every intervention domain contains the support of the corresponding natural responses and confounding is cluster-contained (every*

backdoor path from a cluster member to Y passes through another member of the same cluster), then (i) and (ii) hold, so $\Delta(\Pi) = 0$ for every valid Π satisfying these conditions.

Within this model class the contrapositive reads

$$\Delta(\Pi) > 0 \implies \text{support failure, or} \\ \text{confounding escaping a cluster.}$$

Coarsening need not sacrifice the optimal attainable value: a positive action-space gap can arise from restricted intervention domains or from dependence that a hard whole-cluster intervention destroys, and both premises are tight (App. A). Even when $\Delta(\Pi) = 0$, coarsening may still affect finite-sample regret, optimization dimension, and intervention cost.

Refining the partition: Hierarchical QCBO (HQCBO). The practitioner may declare a *hierarchy* H : per cluster, an optional sub-partition (default: singletons). No new graph is supplied. HQCBO runs Algorithm 1 at Π and, when the incumbent plateaus (trigger parameters in §5), splits the incumbent arm’s clusters via H , refusing any split that fails the validity re-check (Prop. 1). On acceptance the exploration set is rebuilt and unioned with every explored arm, data carry over keyed by arm (Lemma 1), and only new arms receive the standard n_0 initialization points (App. C). We evaluate a single refinement step. This is a value-guided coarse-to-fine descent of the partition lattice, stopping at the coarsest visited partition whose cluster-level MIS contains a value-optimal arm.

Proposition 4 (Fixed-partition regret and its removal by refinement). *Let $\hat{a}_t$ be the incumbent intervention after trial t and $\mu(\hat{a}_t)$ its true mean value. Define*

$$r_t := \mu(\hat{a}_t) - y^* \geq 0, \quad R_T := \sum_{t=1}^T r_t.$$

Since $y^ = V(\Pi_\perp)$ (Lemma 2) and a fixed partition plays only arms in $\mathcal{U}(\Pi)$,*

$$r_t \geq \Delta(\Pi) \quad \forall t, \quad \Delta(\Pi) > 0 \implies R_T \geq T \Delta(\Pi).$$

If refinement reaches a partition Π_t with $\Delta(\Pi_t) = 0$ and the base optimizer’s incumbent is consistent on the finite arm set, then

$$r_t \rightarrow 0 \implies \frac{R_T}{T} \rightarrow 0.$$

The gap is a property of the action space, so no amount of data lowers r_t below $\Delta(\Pi)$. Refinement corrects it by changing the action space itself. We claim no convergence rate and report post-trigger behavior empirically (§5.3). The trigger never observes y^* , so it also fires when $\Delta(\Pi) = 0$. There refinement costs only the new arms’ initialization.

Table 1: One recipe, three families: each base method’s graph-dependent components and their quotient replacements.

Base	Graph dependence	Quotient replacement
CBO	MIS arms; do-calculus priors	cluster MIS of the C-DAG; two-tier C-DAG priors
DCBO	per-slice arms; temporal mechanisms	cluster MIS per slice; cluster mechanisms per slice
MCBO	per-node GP mechanism factorization	joint multi-output cluster mechanisms

Corollary 2 (Phase-scoped invariance). *Let $\Pi_0 \succeq \Pi_1 \succeq \dots$ be the partitions HQCBO visits under hierarchy H . For hypotheses $\mathcal{G}, \mathcal{G}'$,*

$$\mathcal{G}^{\Pi_j} = (\mathcal{G}')^{\Pi_j} \quad \forall j \implies \tau_H(\mathcal{G}) = \tau_H(\mathcal{G}').$$

Splitting a cluster exposes its internal edges at later phases. Anything confined to a never-split cluster stays protected across the whole run. (Proof: App. A.)

3.3 Why not both for free?

Could a quotient method keep its invariance *and* evaluate sub-cluster interventions when they matter? No: sub-cluster behavior is provably not a function of quotient-level inputs. Prop. 7 (§4.3) exhibits two SCMs that agree on every quotient-level input yet disagree on a sub-cluster intervention, so no method whose inputs factor through $(\Pi, \mathcal{G}^\Pi)$, arm-based, dynamic, or model-based, can distinguish them. Refinement is not a loophole but a controlled purchase: each split spends invariance exactly where it refines (Cor. 2).

4 Instantiations: QCBO, QDCBO, and QMCBO

The contract is met by one recipe: replace each graph-coupled component of a base optimizer with its quotient counterpart (Table 1). We instantiate it for the three main families, in increasing order of graph coupling: CBO (the graph prunes arms and builds priors), DCBO (it additionally carries information across time slices), and MCBO (it is the surrogate’s architecture). Figure 2 shows the structural input each family consumes and its quotient. Estimator mechanics live in App. C and E.

4.1 Arm-based: quotient CBO (QCBO)

QCBO uses only cluster-level structure: a partition Π of the manipulable variables $\mathbf{X}$ and a C-DAG $\mathcal{G}^\Pi$ over its clusters and the singleton nodes. Y and every non-manipulable observable stay singletons, and mixing manipulable and non-manipulable variables in

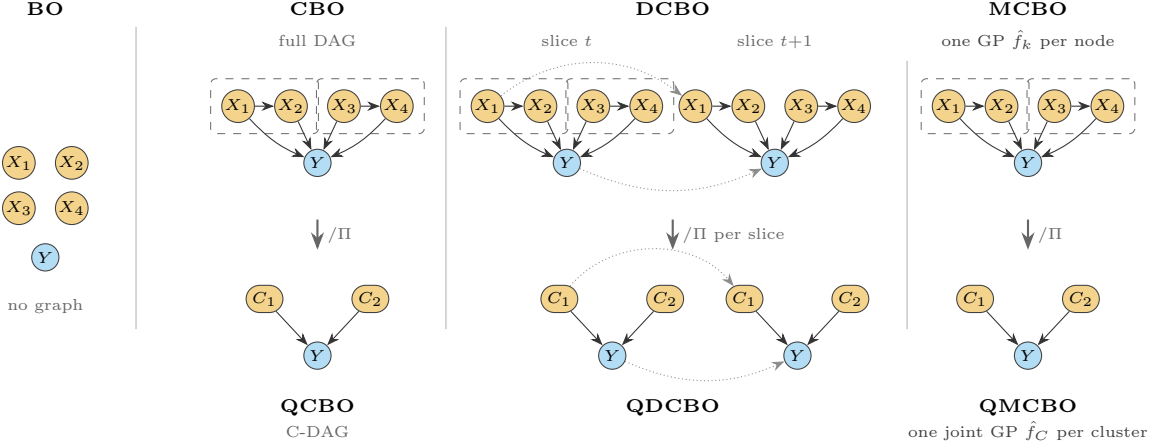


Figure 2: The input contract across the three families. Each base method (CBO, DCBO, MCBO) consumes a full graph. Its quotient counterpart consumes only the partition Π and the C-DAG $\mathcal{G}^\Pi$ ($/\Pi$ arrows). Non-causal BO (left) consumes no graph. Intra-cluster edges — everything inside a dashed box — do not exist at the quotient level, which is the entire content of Theorem 1.

one cluster is disallowed, since intervening on a cluster (jointly setting all its members, $\text{do}(C_k = \mathbf{c}_k)$) must be executable. When the practitioner instead holds a fine-grained hypothesis graph $\mathcal{G}$, $\mathcal{G}^\Pi$ is its quotient after projecting out hidden confounders [Lee and Bareinboim, 2019]. QCBO itself never requires the fine graph. The exploration set is CBO’s own rule (Eq. 2) applied to the quotient, the MIS of $\mathcal{G}^\Pi$ over the manipulable cluster vertices Π_M :

$$\mathbf{ES}^\Pi = \{ U_\Pi(\mathcal{S}) : \emptyset \neq \mathcal{S} \subseteq \Pi_M, \mathcal{S} \subseteq \text{An}_{\mathcal{G}^\Pi}(Y) \}, \quad (3)$$

This is $\text{MIS}(\mathcal{G}^\Pi)$, and $\text{POMIS}(\mathcal{G}^\Pi) \subseteq \text{MIS}(\mathcal{G}^\Pi)$, so the possibly-optimal arms are always retained. Merging manipulable variables coarsens the action space: QCBO can only set a cluster jointly, never a strict subset of it. Algorithm 1 (App. E) summarizes the loop.

Identifiability and the two-tier prior. For each arm $\mathbf{X}_s \in \mathbf{ES}^\Pi$ we pose the cluster-level query $P(Y \mid \text{do}(\mathbf{X}_s))$ on $\mathcal{G}^\Pi$ and run the complete ID algorithm of Shpitser and Pearl [2006] (via `ananke`; Ananke). Identifiability decides the arm’s *prior tier*:

$$m_s(\mathbf{x}_s) = \begin{cases} \hat{\mathbb{E}}[Y \mid \text{do}(\mathbf{X}_s = \mathbf{x}_s)] & P(Y \mid \text{do}(\mathbf{X}_s)) \\ & \text{identified in } \mathcal{G}^\Pi, \\ \bar{Y}_{\text{obs}} & \text{otherwise.} \end{cases} \quad (4)$$

The identifiability test is exact where it applies: ID is complete on the C-DAG, and identification from $\mathcal{G}^\Pi$ is sound for every fine graph in its class [Anand et al., 2023]. A non-identifiable arm learns its response from experimental data alone. Estimator mechanics (the GP plug-in pipeline instantiating identified formulas) are in App. C.

Assumption 1 (Valid C-DAG; estimator boundary). *The directed part of $\mathcal{G}^\Pi$ is acyclic, which constrains only inter-cluster structure (intra-cluster acyclicity can even be relaxed; Madaleno et al., 2026b), coincides with validity of the coarsening when $\mathcal{G}^\Pi$ is a quotient, and is checkable on the cluster-level object. The ID test is complete, while the estimator pipeline behind Eq. (4) is sound but not complete: identifiable queries it misses receive the uninformative tier (in §5 this set is empty).*

Proposition 5 (Identity recovery). *At the identity partition,*

$$\Pi = \Pi_\perp \Rightarrow \mathcal{G}^\Pi = \mathcal{G}^*, \mathbf{ES}^\Pi = \text{MIS}(\mathcal{G}^*), m_s^\Pi = m_s^{\text{CBO}},$$

so QCBO reduces to CBO. (Proof: App. A.)

App. C documents two deviations of the CBO authors’ shipped exploration sets from the MIS definition and the corrected sets both methods use here.

4.2 Dynamic: quotient DCBO (QDCBO)

Dynamic CBO (DCBO; Aglietti et al., 2021) optimizes a sequence of interventions on a dynamic Bayesian network. The graph enters the per-slice exploration sets, the emission and transition estimates, and the propagation of slice t ’s optimum into slice $t+1$ ’s priors. QDCBO applies the recipe per slice. *Graph*: one vertex $C_{k,t}$ per cluster per slice, emission edges quotienting within a slice and transition edges across, validity re-checked per slice (Prop. 1). *Mechanisms*: exploration sets lifted to whole-cluster patterns and emission/transition estimates replaced by joint cluster mechanisms (the coregionalization construction of

QMCBO below), with slice-to-slice propagation as in DCBO at cluster level. *Identity*: singleton clusters reduce every component to stock DCBO.

Corollary 3 (QDCBO inherits the contract). *Every graph-dependent component of QDCBO is a function of $(\Pi, \mathcal{G}^\Pi)$ applied per slice, so Theorem 1 applies, and Prop. 2 applies to its per-slice reachable intervention set.*

4.3 Model-based: quotient MCBO (QMCBO)

MCBO [Sussex et al., 2023] fits one GP per *mechanism*, evaluates interventions by composing these GPs along the DAG, and selects them by GP-UCB: the graph is the surrogate’s architecture. *Graph*: QMCBO composes along the C-DAG. *Mechanisms*: one joint mechanism per cluster, $P(\mathbf{X}_C \mid \mathbf{X}_{\text{pa}(C)})$, a multi-output GP whose coregionalization kernel couples the member coordinates [Bonilla et al., 2007], with joint within-cluster sampling and intervention targets lifted to whole-cluster patterns. *Identity*: singleton clusters reduce to MCBO’s per-node GPs.

Corollary 4 (QMCBO inherits the contract). *Every graph-dependent component of QMCBO is a function of $(\Pi, \mathcal{G}^\Pi)$, so Theorem 1 applies. At $\Pi = \Pi_\perp$, QMCBO is exactly MCBO, and its reachable intervention set (whole-cluster do patterns) equals QCBO’s cluster-level arms, so Prop. 2 applies.*

Proposition 6 (Soundness of cluster mechanisms). *Assume causal sufficiency and a valid Π . For every union-of-clusters intervention $\mathbf{x}_S$,*

$$P(\mathbf{v} \mid \text{do}(\mathbf{x}_S)) = \prod_{C \notin S} P(\mathbf{x}_C \mid \mathbf{x}_{\text{pa}(C)}) \Big|_{\mathbf{x}_S}.$$

(Proof: App. A.)

The mechanism must be joint: per-member single-output GPs are exact only when the members are conditionally independent given the cluster’s parents.

Proposition 7 (Exposure). *There exist SCMs M_1, M_2 , causally sufficient and valid for the same Π , with*

$$\begin{aligned} \mathcal{I}_\Pi(M_1) &= \mathcal{I}_\Pi(M_2), \\ P^{M_1}(Y \mid \text{do}(\mathbf{x}_A)) &\neq P^{M_2}(Y \mid \text{do}(\mathbf{x}_A)) \end{aligned}$$

for a strict subset $A \subsetneq C$ of some cluster, where $\mathcal{I}_\Pi(M)$ collects M ’s quotient-level inputs: the C-DAG, the joint cluster mechanisms, and every union-of-clusters interventional distribution. (Proof: App. A.)

No quotient-respecting method, arm-based, dynamic, or model-based, can therefore evaluate sub-cluster

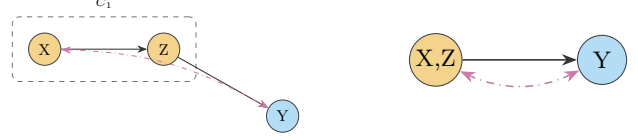


Figure 3: Coarsening can destroy identifiability the fine graph had. ToyGraph’s fine DAG (left): $X \rightarrow Z \rightarrow Y$ with latent confounding $X \leftrightarrow Y$. $\text{do}(Z)$ is identifiable (backdoor through X) and $\text{do}(X)$ is identifiable (front-door through Z). Its quotient under $\Pi = \{\{X, Z\}\}$ (right) is a *bow*: $C_1 \rightarrow Y$ with $C_1 \leftrightarrow Y$, so $P(Y \mid \text{do}(C_1))$ is not identifiable from the C-DAG and the single arm runs on the uninformative prior tier (Eq. 4).

arms from its inputs alone (§3.3). ToyGraph realizes the arm-based face of this trade (Fig. 3): merging $\{X, Z\}$ imports X ’s latent confounding with Y into the single cluster arm, so $P(Y \mid \text{do}(C))$ is not identifiable from the quotient even though both fine queries are identifiable, and the arm runs on the uninformative prior tier (§5.4).

5 Experiments

The experiments answer four questions on three minimal SCMs with closed-form oracles (MINIMALBENCH, App. B), then check that the contract carries across optimizer families on the field’s own benchmarks. (i) Does quotient invisibility give exact invariance? (ii) Can that matter for optimization quality? (iii) What does a too-coarse partition cost? (iv) Does the contract transfer to CBO, DCBO, and MCBO? Each experiment reports five items: the SCM, the perturbation, the oracle, the prediction, and the result.

Protocol. Every condition holds the true SCM, the observational data ($n_{\text{obs}} = 100$ per run, one shared dataset per SCM), y^* , seeds, and trials fixed, and injects the misspecification only into the assumed structure the method reasons with (App. C). The methods share one estimator backend and differ only in the structure they consume: CBO identifies on the full assumed DAG over its MIS arms (by Prop. 5 it is QCBO at the finest partition), QCBO runs the same algorithm on the quotient. 30 seeds, $T=50$ trials (60 on MEDIATEDCHAIN), 3 initial interventional points per arm, unit costs. HQCBO’s plateau trigger fires when the incumbent improves by less than $\delta=10^{-3}$ over $k=5$ trials, tuned once and fixed. Behavioral robustness is scored paired-by-seed: $\Delta_t = \text{metric}(\text{misspecified}, s, t) - \text{metric}(\text{correct}, s, t)$, and $\sup |\Delta|$ over trials and seeds is the invariance check. Table 2 classifies every perturbation in advance.

Table 2: Perturbation taxonomy of the minimal suite, emitted from the same structural module the methods consume. By Theorem 1, rows with an unchanged quotient are predicted invariant. The final column reports the measured paired deviation $\sup_{s,t} |\Delta|$ for QCBO over 30 seeds. PP = PARALLEL PARENT, FD = FRONT DOOR.

Cond.	Edit (assumed graph)	Quotient changed	Fine MIS changed	Fine prior changed	$\sup \Delta $
A1 (PP)	del $X_1 \rightarrow Y$ (quotient-redundant)	$\times$	$\checkmark$	—	0
A2 (PP)	add $X_1 \rightarrow X_2$ (intra-cluster bow)	$\times$	$\times$	$\checkmark$	0
A3 (PP)	assume $X_1 \leftrightarrow Y$ (quotient-visible)	$\checkmark$	$\times$	$\checkmark$	21.4
B1 (FD)	assume $X_1 \leftrightarrow M$ (intra-cluster conf.)	$\times$	$\times$	$\checkmark$	0

5.1 Experiment A: exploration-set corruption

SCM: PARALLEL PARENT (Fig. 1; $\lambda=1$), fine MIS $\{X_1\}, \{X_2\}, \{X_1, X_2\}$, $y^* = 0$ at the joint arm. The partition $C_1 = \{X_1, X_2\}$ has $\Delta(\Pi) = 0$. *Perturbation*: A1 deletes $X_1 \rightarrow Y$ (quotient-redundant), A2 adds the intra-cluster $X_1 \rightarrow X_2$, and A3 assumes a spurious $X_1 \leftrightarrow Y$ (quotient-visible negative control). *Oracle*: under A1 the fine MIS collapses to $\{X_2\}$, whose best reachable value is $\lambda(4 + \text{Var}X_1) = 4.25$. *Prediction*: CBO loses the optimal arm structurally under A1. QCBO is invariant under A1 and A2 but not under A3. *Result*: CBO pins at 4.25 on every seed (Fig. 4a). QCBO’s arm choices, intervention values, and incumbents are identical under A0 and A1 on every seed ($\sup |\Delta| = 0$, Table 2) and it reaches y^* . Under A3 the C-DAG gains $C_1 \leftrightarrow Y$, the cluster arm drops to the uninformative tier, and the trajectory moves off its A0 path on every seed. Under A2 even fine CBO moves by $\sup |\Delta| < 10^{-2}$: the corrupted singleton is irrelevant to the optimum, which motivates Experiment B’s design.

5.2 Experiment B: prior corruption without arm loss

SCM: FRONT DOOR: $X_1 \rightarrow M \rightarrow Y$ with latent $X_1 \leftrightarrow Y$ and a *narrow-well* target, $y^* = 0$ at $\text{do}(M=1)$. The well is a small fraction of the search domain, so the do-calculus prior’s head start decides whether the optimum is sampled immediately or found by global search (App. B). *Perturbation*: B1 assumes a spurious intra-cluster confounder $X_1 \leftrightarrow M$. *Oracle*: the fine MIS is untouched, but both fine arms lose identification and drop to the uninformative tier (Eq. 4). *Prediction*: transient damage for fine CBO, exact invariance for QCBO, since the edit is intra-cluster. *Result*: fine CBO under B1 is slower early (Fig. 4b), paying $\Delta \hat{R}_{50} = 6.1 \pm 1.0$ cumulative regret (paired, mean $\pm$ s.e., 30 seeds), yet recovers essentially the same final value (0.006 vs 0.001, against an observational baseline of 1.64). QCBO’s trajectories, arm choices,

and intervention values are exactly identical across the two conditions ($\sup |\Delta| = 0$), its single cluster arm on the uninformative tier in both.

5.3 Experiment C: the price of coarsening, and buying it back

SCM: MEDIATED CHAIN: $X_1 \rightarrow X_2 \rightarrow Y$ with $Y = (X_2 - 4)^2$, $X_2 = 2X_1$ (plus noise), and the *policy box* $D(X_2) = [-2, 2]$, fine MIS $\{X_1\}, \{X_2\}$ (on a chain the joint is not minimal, Lemma 1). *Perturbation*: none; the treatment is the partition $C_1 = \{X_1, X_2\}$ itself. *Oracle*: $V(\Pi) = 4$, $y^* = 0.04$, $\Delta(\Pi) = 3.96$: the fine optimum $\text{do}(X_1=2)$ drives X_2 ’s natural response outside the box, failing premise (i) of Prop. 3, and the whole-cluster arm must clamp X_2 into the box. *Prediction*: fixed-partition cumulative regret grows at rate $\Delta(\Pi)$ (Prop. 4), and refinement removes it. *Result*: CBO descends to 0.040; fixed-partition QCBO pins at 4.000 on every seed with $\hat{R}_{60} = 252.7 \pm 1.8$; HQCBO triggers at trial 7.5 ± 0.1 (30/30 seeds, no refused splits), splits $\{X_1, X_2\}$, and descends to 0.040 ± 0.000 with $\hat{R}_{60} = 46.7 \pm 2.0$, against CBO’s 10.4 ± 3.0 (Fig. 5). Phase 1 is byte-identical to QCBO’s prefix on every seed (Cor. 2).

5.4 Experiment D: portability across families

Setup: each quotient method against its base on the experiments the base method’s own paper ran, in that paper’s own metric, through the authors’ released stacks: CBO vs QCBO on the 2020 paper’s Toy, Synthetic, and coral-reef examples (best $E[Y \mid \text{do}]$, 10 seeds); DCBO vs QDCBO on its stationary, independent, and nonstationary DBNs at $T=3$ (20 seeds); MCBO vs QMCBO on its ToyGraph and PSAGraph tasks (5 seeds; the paper’s remaining problems are function networks, where a partition of function nodes has no causal reading). DAGs and quotients are drawn in App. D. Fig. 6 overlays the trajectories and Table 3 reports finals and realized prices.

Value. The quotient matches or beats its base on six of the eight experiments. QMCBO essentially reaches

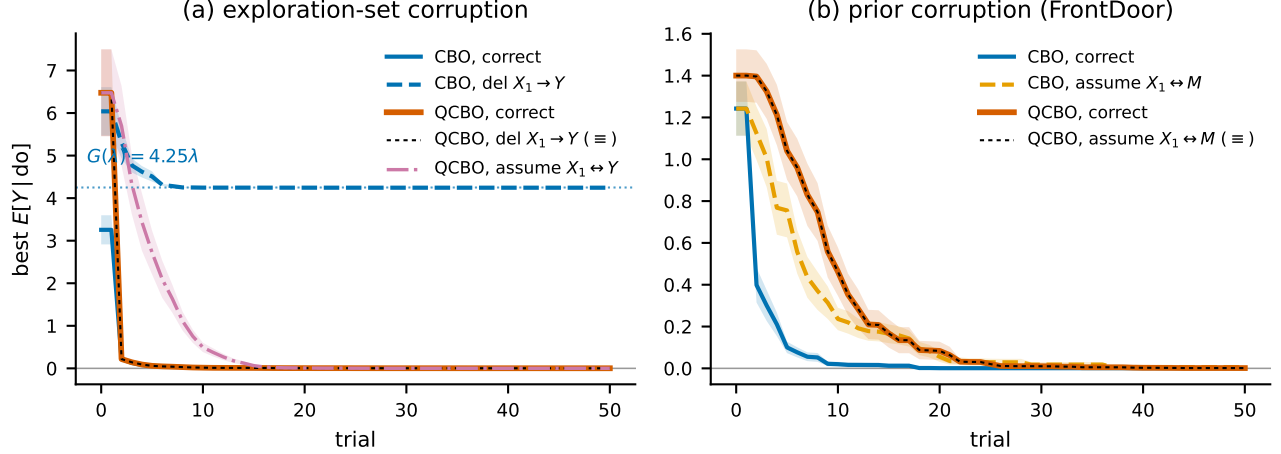


Figure 4: Correct (solid) vs misspecified (dashed) trajectories, mean $\pm$ s.e. over 30 seeds. (a) Exploration-set corruption (PARALLELPARENT): wrong-fine CBO plateaus at the analytic gap $G(\lambda) = 4.25\lambda$. QCBO’s misspecified trajectory coincides exactly with its correct one (black dashes over orange, $\sup |\Delta| = 0$). The quotient-visible control (pink) moves QCBO off its path. (b) Prior corruption (FRONTDOOR): the assumed intra-cluster $X_1 \leftrightarrow M$ costs fine CBO its do-calculus head start — transiently slower, same final value — while QCBO is exactly invariant.

ToyGraph’s optimum (2.150 ± 0.013 vs 2.172) where MCBO’s recorded best-so-far trajectory remains flat across the 100 BO iterations (1.14 ± 0.27). QMCBO models within-cluster correlation jointly (Prop. 6), a difference consistent with, though not by itself shown to cause, the gap. On PSAGraph both converge to the same plateau (-5.15 ± 0.00), QMCBO in $3.0\times$ less wall-clock. The two material positive prices are exactly where Prop. 2 predicts them: Synthetic, whose optimum $\text{do}(B, D)$ splits the domain partition, and, mildly, Coral. On Toy the single coarse arm is the bow of Fig. 3. QCBO keeps it on the uninformative prior and still matches CBO (-2.13 ± 0.03 vs -2.16 ± 0.00).

Invariance within each family. Under within-family quotient-invisible perturbations of the model’s view, QMCBO’s trajectories are identical to their correct-graph runs on 10/10 (env, seed) pairs while MCBO’s shift on 8/10, in a direction that is not predictable (on these SCMs the wrong graph sometimes *helped*). QDCBO is likewise invariant on its three setups under the intra-slice edit while DCBO moves (App. F).

6 Related work

QCBO occupies an unfilled cell in the CBO landscape: it neither assumes the full DAG nor learns one, but takes a coarser, more reliable input and identifies on the quotient (Table 5, App. F). Methods relaxing the full-graph requirement learn structure online at per-trial cost (GACBO maintains a DAG posterior

Table 3: Realized price of coarsening per family, each pair on the base method’s own experiment suite (final Y , mean $\pm$ s.e.). Price is signed so that positive means the quotient did worse in the task’s direction.

Family	Experiment	Base final	Quotient final	Price	Task
CBO vs QCBO	Toy	-2.16 ± 0.00	-2.13 ± 0.03	$+0.03$	min
CBO vs QCBO	Synthetic	-3.14 ± 0.19	-1.25 ± 0.07	$+1.89$	min
CBO vs QCBO	Coral	36.08 ± 0.02	36.42 ± 0.01	$+0.34$	min
DCBO vs QDCBO	stat ($T=3$)	-6.14 ± 0.05	-6.43 ± 0.02	-0.29	min
DCBO vs QDCBO	ind ($T=3$)	-3.12 ± 0.06	-5.57 ± 0.15	-2.45	min
DCBO vs QDCBO	nonstat ($T=3$)	8.03 ± 0.90	6.33 ± 0.29	-1.69	min
MCBO vs QMCBO	ToyGraph	1.14 ± 0.27	2.15 ± 0.01	-1.01	max
MCBO vs QMCBO	PSAGraph	-5.15 ± 0.00	-5.15 ± 0.00	-0.00	max

[Mukherjee et al., 2024], CEO hedges a graph posterior through an entropy-based acquisition [Branchini et al., 2023], Durand et al. [2025] identify $\text{Pa}(Y)$ under a Y -leaf assumption) or assume a known equivalence class [Park et al.]. Averaging only *dilutes* a wrong edge in proportion to posterior mass, whereas the quotient pays nothing per trial and is exactly invariant to any edit it does not resolve. The identification machinery we rely on is established (C-DAG do-calculus [Anand et al., 2023], C-DMGs over ADMGs [Ferreira and Assaad, 2025], cluster-level Markov properties [Wahl et al., 2024], relaxed admissibility [Bytez.com et al., 2025], C-DAGs as discovery background knowledge [Padh and Kilbertus]), but none of it performs optimization. Closest in spirit, Zeitler [2025] coarsen the *outcome* as a cheap proxy [Poloczek et al., 2017, Beckers and Halpern, 2019]. Causal abstraction relates decisions across levels of granularity, and the quotient is such a map, applied to the manip-

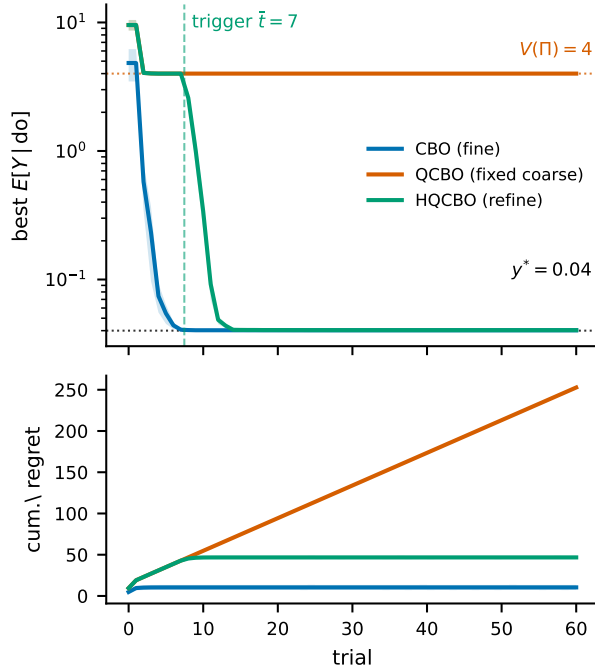


Figure 5: The price of coarsening and its recovery (MEDIATEDCHAIN, mean $\pm$ s.e., 30 seeds). Top: best-so-far against the analytic $V(\Pi) = 4$ and $y^* = 0.04$ (log scale). The vertical line marks the mean refinement trigger. Bottom: cumulative regret — linear growth for the fixed partition, flattening for HQCBO, with fine CBO the best constant.

ulable *inputs*.

7 Discussion and limitations

Theorem 1 is structural: coarsening acts as a hard regularizer on the hypothesis space, trading resolution (Prop. 2) for immunity to a class of misspecifications. Caveats. (i) The invariance belongs to the *quotient*, and the guarantee is about *behavior*. A wrong quotient produces the same trajectory for the wrong problem. (ii) DCBO and MCBO cannot join the fixed-objective seam (per-environment coupling), so QDCBO and QMCBO are each evaluated within their own family’s harness with the base method’s analytic objective held fixed. (iii) Faithfulness does not transfer to the cluster level [Wahl et al., 2024]. (iv) Unit intervention costs only. (v) Building the exploration set costs one MIS test and one complete-ID check per candidate cluster union, up to $2^K - 1$ of each on the quotient: exponential in clusters, not in variables. (vi) QMCBO assumes causal sufficiency (Prop. 6), as MCBO does. A rollback rule bounding the lossless-trigger tax, and discovering the hierarchy from the run’s own data, are natural follow-ups.

8 Conclusion

Coarsening runs causal Bayesian optimization on a coarser, more reliable structural commitment, with provable invariance to intra-cluster misspecification, an action-space gap that is zero under the conditions of Prop. 3, and refinement to correct the gap where it is positive (Prop. 4). The practitioner dials resolution against robustness by choosing how much structure to commit to, in any of the three families.

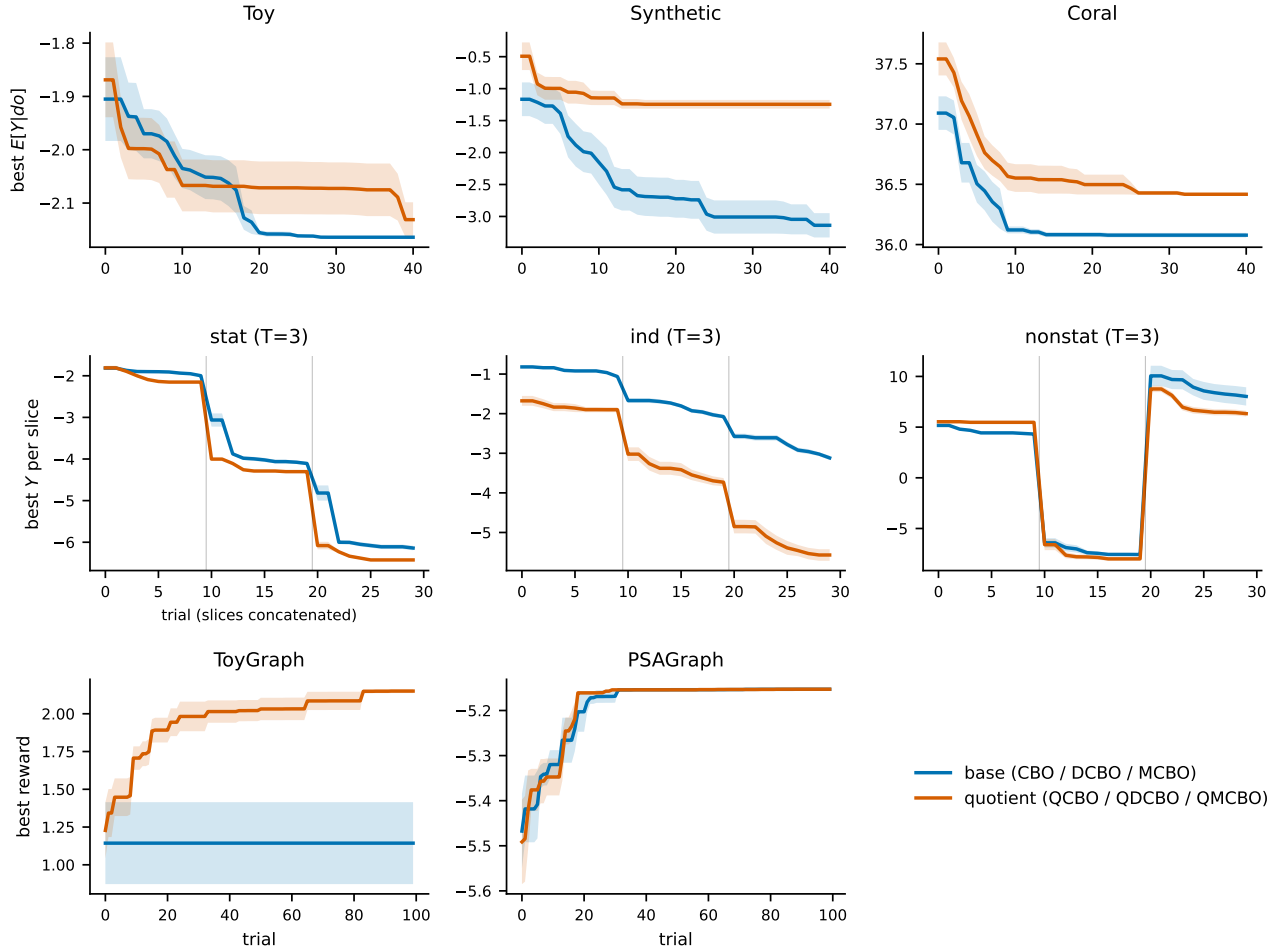


Figure 6: Each quotient method against its base on the base paper’s own experiments and metric (mean $\pm$ s.e.). Top: CBO vs QCBO (best $\mathbb{E}[Y | do]$, minimization). Middle: DCBO vs QDCBO (best Y per time slice, trials concatenated across slices). Bottom: MCBO vs QMCBO (best reward, maximization).

A Proofs

Notation is as in §4.1 and §3.

Proof of Theorem 1. Write $\mathcal{A}$'s run as a sequence of decisions $d_1, d_2, \dots$ (which arm/action to evaluate, model refits, stopping). By the factoring hypothesis each d_t is a deterministic function $d_t = f_t(Q_\Pi(\mathcal{G}), \mathcal{D}_{t-1}, \omega_t)$ of the quotient pair, the data collected so far, and the RNG stream. Under $\mathcal{G}$ and $\mathcal{G}'$ with $Q_\Pi(\mathcal{G}) = Q_\Pi(\mathcal{G}')$ the first arguments coincide. The objective (the true SCM) and the seeds coincide by the fixed-objective protocol, so $\mathcal{D}_0, \omega_1$ coincide, hence d_1 coincides, hence $\mathcal{D}_1$ coincides. Induction over t gives equality of the entire trajectory. The scope claimed after the theorem follows because the latent projection removes only latent vertices and an edge between two observables is never an interior vertex of a projection path, so $(\mathcal{G}')^*$ and $\mathcal{G}^*$ differ exactly by the modified intra-cluster edges; the quotient keeps a directed (resp. bidirected) cluster-level edge iff some fine edge crosses two distinct clusters, so edits confined within clusters, or redundant across them, leave $Q_\Pi(\mathcal{G})$ unchanged. $\square$

Proof of Prop. 6. Under causal sufficiency the fine SCM is Markov relative to $\mathcal{G}$ with mutually independent exogenous noises, so for any intervention $\text{do}(\mathbf{x}_S)$ on a union of clusters S the fine truncated factorization gives $P(\mathbf{V} \mid \text{do}(\mathbf{x}_S)) = \prod_{k \notin S} P(X_k \mid \mathbf{X}_{\text{pa}_{\mathcal{G}}(k)})|_{\mathbf{x}_S = \mathbf{x}_S}$. Group the factors by the clusters of Π . For a cluster $C \not\subseteq S$, the product of its members' factors is, by the chain rule applied in any topological order of $\mathcal{G}$ restricted to C ,

exactly the joint conditional $P(\mathbf{X}_C \mid \mathbf{X}_{\text{pa}(C)})$, where $\text{pa}(C)$, the union of C 's parent clusters, contains every out-of-cluster variable appearing in a member's conditioning set (validity of Π guarantees the quotient is acyclic, so $\text{pa}(C)$ is well defined and precedes C in C-DAG order). The fine factorization therefore equals the product over clusters of the joint cluster mechanisms with the clusters of S clamped at $\mathbf{x}_S$: the cluster-level truncated factorization. Note both uses of sufficiency: independent noises across clusters (no bidirected edges) and the absence of unobserved common causes that would make a member's conditional depend on variables outside $\text{pa}(C) \cup C$. Replacing $P(\mathbf{X}_C \mid \mathbf{X}_{\text{pa}(C)})$ by the product of its per-member marginals would be exact iff the members are conditionally independent given $\text{pa}(C)$, in particular when C contains no intra-cluster edges — hence the joint mechanism in the construction. $\square$

Proof of Prop. 7. Take $\mathbf{X} = \{X_2, X_3\}$ as one root cluster C , target $Y = X_2 + \varepsilon_Y$ its child, all noises standard normal and independent (sufficiency holds, and any Π clustering $\{X_2, X_3\}$ is valid). Let M_1 : $X_2 = \varepsilon_2$, $X_3 = aX_2 + \varepsilon_3$; and M_2 : $X_3 = \varepsilon'_3$, $X_2 = a'X_3 + \varepsilon'_2$, with $a' = a/(1+a^2)$, $\text{Var}(\varepsilon'_3) = 1+a^2$, and $\text{Var}(\varepsilon'_2) = 1-a^2/(1+a^2)$. Both induce the same zero-mean Gaussian joint on (X_2, X_3) , with covariance $\begin{pmatrix} 1 & a \\ a & 1+a^2 \end{pmatrix}$, hence the same joint cluster mechanism (here, the same root-cluster distribution), and the same $P(Y \mid \text{do}(\mathbf{x}_C))$ for the whole-cluster intervention (both set X_2 directly). But under the sub-cluster intervention $\text{do}(X_3 = x)$: in M_1 , $X_2 = \varepsilon_2$ is unaffected, so $\mathbb{E}[Y \mid \text{do}(X_3=x)] = 0$ for all x ; in M_2 , $X_2 = a'x$, so $\mathbb{E}[Y \mid \text{do}(X_3=x)] = a'x \neq 0$ for $x \neq 0$, $a \neq 0$. Every quotient-respecting input (the C-DAG, the joint cluster mechanisms, all whole-cluster interventional distributions) coincides between M_1 and M_2 , yet the sub-cluster query differs. No function of those inputs can answer it. $\square$

Proof of Prop. 5. Singleton clusters make the quotient map the identity on $\mathcal{G}^*$, so $\mathcal{G}^\Pi = \mathcal{G}^*$, unions of singleton clusters are exactly the subsets of $\mathbf{X}$, and the MIS condition of Eq. (3) coincides verbatim with Eq. (2). The prior construction poses the same do-calculus queries CBO poses on the full graph, so priors coincide arm-for-arm. $\square$

Lemma 2 (MIS reduction). *Let G be an ADMG with manipulable vertices M and target Y , let $\emptyset \neq \mathbf{W} \subseteq M$, and set $\mathbf{W}' = \mathbf{W} \cap \text{An}(Y)$ computed in $G_{\mathbf{W}}$. Then (i) $P(Y \mid \text{do}(\mathbf{W} = \mathbf{w})) = P(Y \mid \text{do}(\mathbf{W}' = \mathbf{w}|_{\mathbf{W}'}))$ for every $\mathbf{w}$; (ii) $\mathbf{W}'$ is a MIS (or empty, when no member reaches Y); and (iii) consequently $\min_{\mathbf{X}_s \in \text{MIS}(G)} \mu^*(\mathbf{X}_s) = \min_{\emptyset \neq \mathbf{W} \subseteq M} \mu^*(\mathbf{W})$: restricting the search to MIS loses no value.*

Proof. (i) In $G_{\mathbf{W}}$ the members of $\mathbf{W} \setminus \mathbf{W}'$ have no directed path to Y , so Y is unaffected by their values under the intervention; dropping them from the intervention changes no mechanism on any directed path into Y (do-calculus rule 3). (ii) A directed path from $w' \in \mathbf{W}'$ to Y in $G_{\mathbf{W}}$ passes through no intervened vertex (their incoming edges are cut), hence through no member of $\mathbf{W} \supseteq \mathbf{W}'$; the same path exists in $G_{\mathbf{W}'}$, which only restores edges, so $\mathbf{W}' \subseteq \text{An}(Y)$ there. (iii) By (i) and (ii) every candidate's value is attained by a MIS, and $\text{MIS}(G)$ is a subset of the candidates. $\square$

Proof of Prop. 1. Take $\mathbf{X} = \{a_1, a_2, b_1, b_2\}$ with fine edges $a_1 \rightarrow b_1$ and $b_2 \rightarrow a_2$. The coarsest partition $\Pi = \{\{a_1, a_2, b_1, b_2\}\}$ is valid: both edges are intra-cluster, so the quotient has a single manipulable node and is acyclic. Its refinement $\Pi' = \{\{a_1, a_2\}, \{b_1, b_2\}\}$ exposes both edges as inter-cluster ($a_1 \rightarrow b_1$ contributes $A \rightarrow B$ and $b_2 \rightarrow a_2$ contributes $B \rightarrow A$), inducing the 2-cycle $A \rightleftarrows B$, so $\mathcal{G}^{\Pi'}$ is cyclic and Π' is invalid. Decidability is immediate from the acyclicity clause of Def. 1. $\square$

Proof of Lemma 1. (i) Membership at Π' is the stated hypothesis (any union of Π -clusters is a union of Π' -clusters, so the candidate exists at Π' ; being a MIS of $\mathcal{G}^{\Pi'}$ puts it in $\mathbf{ES}^{\Pi'}$). For the identification functional: quotienting is compositional, $\mathcal{G}^\Pi$ being the quotient of $\mathcal{G}^{\Pi'}$ under the induced partition of Π' -clusters, so every fine graph compatible with $\mathcal{G}^{\Pi'}$ is compatible with $\mathcal{G}^\Pi$. Identifiability from $\mathcal{G}^\Pi$ means a single functional of the observational distribution equals $\mathbb{E}[Y \mid \text{do}(\mathbf{X}_s)]$ across all SCMs whose graph is compatible with $\mathcal{G}^\Pi$ [Anand et al., 2023]. A fortiori the same functional works across the smaller class compatible with $\mathcal{G}^{\Pi'}$. (ii) Non-monotone membership is witnessed by ToyGraph: $X \cup Z$ is the single MIS of the coarse quotient, while $\{X, Z\}$ is not a MIS at $\Pi_\perp$ — mutilating Z cuts $X \rightarrow Z$, leaving X causally inert, so $\text{do}(X, Z) \equiv \text{do}(Z)$. $\square$

Proof of Prop. 2. By Lemma 2(iii) applied to $\mathcal{G}^\Pi$, $\min_{A \in \mathbf{ES}^\Pi} \mu^*(A)$ equals the minimum of μ^* over all of $\mathcal{U}(\Pi)$ (the value μ^* of a cluster union is a property of the underlying SCM, the same at every partition that can express the union), which is $V(\Pi)$ by definition. Unions of Π -clusters are unions of Π' -clusters and, at $\Pi_\perp$, all non-empty

subsets of $\mathbf{X}$: $\mathcal{U}(\Pi) \subseteq \mathcal{U}(\Pi') \subseteq \mathcal{U}(\Pi_\perp)$. Minima over nested families are ordered, $V(\Pi_\perp) \leq V(\Pi') \leq V(\Pi)$, which gives $0 \leq \Delta(\Pi') \leq \Delta(\Pi)$ (in particular $\Delta(\Pi) \geq 0$). Equality $\Delta(\Pi) = 0$ holds iff the unrestricted minimum is attained by a union of Π -clusters, and by Lemma 2(iii) at $\Pi_\perp$ the unrestricted minimum is min over Eq. (1)'s exploration set: exactly the stated condition. $\square$

Proof of Prop. 4. By Lemma 2(iii) at $\Pi_\perp$, $y^* = V(\Pi_\perp)$. A fixed partition only ever executes arms in $\mathcal{U}(\Pi)$ (Eq. 3), so the incumbent satisfies $\mu(\hat{a}_t) \geq V(\Pi)$ at every t , hence $r_t = \mu(\hat{a}_t) - y^* \geq V(\Pi) - V(\Pi_\perp) = \Delta(\Pi)$ and $R_T \geq T\Delta(\Pi)$. If some visited Π_t has $\Delta(\Pi_t) = 0$, its exploration set contains an arm of value y^* (Prop. 2); consistency of the incumbent on the finite arm set then gives $\mu(\hat{a}_t) \rightarrow y^*$, i.e. $r_t \rightarrow 0$, and Cesàro averaging gives $R_T/T \rightarrow 0$. $\square$

Proof of Prop. 3. Fix a fine intervention $A \subseteq \mathbf{X}$ and $a \in D(A)$, and let $\text{cl}_\Pi(A)$, $B = \text{cl}_\Pi(A) \setminus A$, and B_a be as in the statement. Since Π partitions the manipulable variables, $\text{cl}_\Pi(A) \in \mathcal{U}(\Pi)$. By the consistency of interventions, setting B to the value it would have taken changes nothing realization by realization: $Y_a = Y_{a,B_a}$. Premise (i) makes every realized b feasible for the joint intervention on $\text{cl}_\Pi(A)$, and premise (ii) lets the expectation factorize into a mixture over feasible joint interventions,

$$\mathbb{E}[Y_a] = \int \mathbb{E}[Y_{a,b}] dP(B_a = b) \geq \inf_{b \in D_{B(a)}} \mathbb{E}[Y_{a,b}] \geq \mu^*(\text{cl}_\Pi(A)) \geq V(\Pi).$$

Minimizing the left side over $a \in D(A)$ and then over $A \subseteq \mathbf{X}$ gives, by Lemma 2(iii) at $\Pi_\perp$, $V(\Pi_\perp) \geq V(\Pi)$. The reverse inequality holds because the coarse intervention family is contained in the fine family ($\mathcal{U}(\Pi) \subseteq \mathcal{U}(\Pi_\perp)$, Prop. 2), so $V(\Pi) = V(\Pi_\perp)$ and $\Delta(\Pi) = 0$. This treats an arbitrary A , possibly meeting several clusters, in one step: the closure and B_a are joint objects, and no per-cluster iteration is needed. $\square$

Proof of Cor. 1. Represent the semi-Markovian SCM with mutually independent exogenous variables $\mathbf{U}$, every latent common cause a single shared exogenous variable (a bidirected edge in $\mathcal{G}^*$). Full intervention support gives premise (i) directly. For premise (ii): under $\text{do}(A = a)$, B_a is a measurable function of $\mathbf{U}_B$, the exogenous variables entering the mechanisms of B and of its non-intervened ancestors. At fixed (a, b) , $Y_{a,b}$ is a measurable function of $\mathbf{U}_Y$, the exogenous variables entering the mechanisms of Y and of its ancestors outside $\text{cl}_\Pi(A)$ (all of $\text{cl}_\Pi(A)$ is intervened in this world, so no path into Y passes through it). Suppose some $U^* \in \mathbf{U}_B \cap \mathbf{U}_Y$. Then U^* has a directed path into some member of B — take b^* to be the *first* $\text{cl}_\Pi(A)$ -member on that path — and a directed path into Y avoiding $\text{cl}_\Pi(A)$ entirely. Concatenating them tail-to-tail yields a backdoor path from b^* to Y whose interior avoids $\text{cl}_\Pi(A)$, hence avoids b^* 's own cluster: a backdoor path escaping the cluster, which cluster-contained confounding excludes. So $\mathbf{U}_B$ and $\mathbf{U}_Y$ are disjoint sets of mutually independent exogenous variables and $B_a \perp Y_{a,b}$ for every fixed (a, b) .

Both premises are tight; dropping either admits an SCM with strictly positive gap. *Support, realized by MEDICATEDCHAIN (§5.3).* The policy box restricts hard interventions on X_2 to $[-2, 2]$ while the mechanism $X_2 = 2X_1$ drives X_2 's natural response to 4, the bowl center of $Y = (X_2 - 4)^2$, when $X_1=2$ is set. The best component b lies outside the interventional domain, so $\text{do}(X_1=2)$ strictly beats every in-box whole-cluster intervention: $\Delta(\Pi) = 3.96$ (App. B). *Independence.* Let $C = \{A, B\}$ with latent U , mechanisms $B = U$ and $Y = -BU + (A - 1)^2$, U standard normal. Under $\text{do}(A=1)$ with B free, $\mathbb{E}[Y] = -\mathbb{E}[U^2] = -1$; under any whole-cluster $\text{do}(A=a, B=b)$, $\mathbb{E}[Y] = -b\mathbb{E}[U] + (a - 1)^2 \geq 0$. Intervening severs the exploitable dependence between B and Y : the backdoor path $B \leftrightarrow Y$ escapes the cluster and the gap is 1 despite unconstrained domains. $\square$

Proof of Cor. 2. Induct over phases. Within phase j , every graph-dependent component factors through $(\Pi_j, \mathcal{G}^{\Pi_j})$, and the trigger time and split decision are functions of the trajectory and the declared hierarchy alone. If $\mathcal{G}^{\Pi_j} = (\mathcal{G}')^{\Pi_j}$ and the phase- j prefixes coincide, Theorem 1 applied at Π_j makes the phase- j runs identical, hence the same trigger time and the same refined partition Π_{j+1} . Agreement at every visited Π_j therefore gives $\tau_H(\mathcal{G}) = \tau_H(\mathcal{G}')$. For the scope claim: an edit confined to a never-split cluster is invisible in every $\mathcal{G}^{\Pi_j}$ (the cluster is intact at each phase), hence protected across the whole run; an edit inside a split cluster is invisible before the split but generally visible after (its endpoints then lie in distinct clusters), which is exactly the exposure purchased. Quotient edges between untouched clusters are the same at every phase. $\square$

B MinimalBench: specification and oracles

The three SCMs share one parameterization — Gaussian roots, manipulable domains $[-3, 3]$ (except the MEDIATEDCHAIN policy box), unit intervention costs, target noise $\sigma_Y = 0.1$, no clipping — so every arm value is closed-form. The structural module that defines them (`ccbo/minibench.py`) is the same one the methods, the perturbation registry, the structural tests, and Table 2’s emitter consume, so specification and experiment cannot drift. Observational datasets (5,000 rows, seed 0; every run uses the first $n_{\text{obs}}=100$) are produced by released generators whose built-in oracle gates assert the Monte-Carlo arm values against the closed forms below (tolerance 0.05) at build time.

PARALLEL PARENT (Experiment A). Latent U ; manipulable X_1, X_2 ; target Y :

$$U = 0.4\epsilon_0, \quad X_i = U + 0.3\epsilon_i, \quad Y = \lambda(X_1 - 2)^2 + (X_2 + 2)^2 + \sigma_Y\epsilon_Y,$$

with $\lambda = 1$; the latent projection carries $X_1 \leftrightarrow X_2$ (corr = 0.64), and $\text{Var}X_i = 0.25$. Coarse partition $C_1 = \{X_1, X_2\}$. Closed forms: $\mathbb{E}[Y \mid \text{do}(x_1, x_2)] = \lambda(x_1 - 2)^2 + (x_2 + 2)^2$, so $y^* = 0$ at the joint arm; $\mathbb{E}[Y \mid \text{do}(x_1)] = \lambda(x_1 - 2)^2 + 4.25$ (the free co-parent’s bowl plus its variance) and $\mathbb{E}[Y \mid \text{do}(x_2)] = 4.25\lambda + (x_2 + 2)^2$. Under A1 (delete $X_1 \rightarrow Y$) the fine MIS is $\{X_2\}$ alone — $X_1 \notin \text{An}(Y)$ in every mutilation — so the best reachable fine value is $G(\lambda) = \lambda(4 + 0.25) = 4.25\lambda$, the analytic gap of §5.1. Conditions: A1 = del $X_1 \rightarrow Y$ (quotient-redundant); A2 = add $X_1 \rightarrow X_2$ (intra-cluster bow with the latent $X_1 \leftrightarrow X_2$: $\text{do}(X_1)$ loses identification, membership intact); A3 = assume $X_1 \leftrightarrow Y$ (quotient-visible: the C-DAG gains $C_1 \leftrightarrow Y$ and the cluster arm drops to the uninformative tier; QCBO keeps running).

FRONT DOOR (Experiment B). Latent U ; manipulable X_1, M ; target Y :

$$U = 0.4\epsilon_0, \quad X_1 = U + 0.3\epsilon_1, \quad M = 2X_1 + 0.2\epsilon_M, \\ Y = 2(1 - e^{-(M-1)^2/(2 \cdot 0.3^2)}) + U + \sigma_Y\epsilon_Y.$$

The latent projection carries $X_1 \leftrightarrow Y$. The target is a narrow well (width 0.3 over the 6-wide domain): prior corruption is a global-search phenomenon, and on a smooth quadratic the same class of corruption measurably costs nothing, since any BO self-corrects within a couple of trials (cf. condition A2 in §5.1). Fine MIS = $\{X_1\}, \{M\}$ (chain); $\text{do}(M)$ is identified by backdoor through X_1 , $\text{do}(X_1)$ by the front-door functional through M . Gaussian smoothing gives the closed forms: $\mathbb{E}[Y \mid \text{do}(m)] = 2(1 - e^{-(m-1)^2/0.18})$, so $y^* = 0$ at $\text{do}(M=1)$; $\mathbb{E}[Y \mid \text{do}(x_1)] = 2(1 - \frac{0.3}{\sqrt{0.3^2 + 0.2^2}} e^{-(2x_1-1)^2/(2(0.3^2+0.2^2))})$, best ≈ 0.336 at $x_1 = 1/2$; observational mean $\mathbb{E}[Y] \approx 1.637$. The coarse partition $\{X_1, M\}$ quotients to the bow $C_1 \rightarrow Y$, $C_1 \leftrightarrow Y$: the cluster arm runs on the uninformative tier in *every* condition, and setting M directly makes the partition lossless ($V(\Pi) = y^* = 0$). Condition B1 assumes a spurious intra-cluster confounder $X_1 \leftrightarrow M$: both fine arms lose identification (the backdoor set $\{X_1\}$ for $\text{do}(M)$ is invalidated by the induced collider path, the front-door premise for $\text{do}(X_1)$ by the confounded $X_1 \rightarrow M$ edge), while the quotient — which drops intra-cluster bidirected edges — is unchanged.

MEDIATED CHAIN (Experiment C). No latents; manipulable X_1, X_2 ; target Y :

$$X_1 = 0.5\epsilon_1, \quad X_2 = 2X_1 + 0.2\epsilon_2, \quad Y = (X_2 - 4)^2 + \sigma_Y\epsilon_Y,$$

with $D(X_1) = [-3, 3]$ and the *policy box* $D(X_2) = [-2, 2]$. Fine MIS = $\{X_1\}, \{X_2\}$: the joint is not minimal on a chain (mutilating X_2 severs X_1 from $\text{An}(Y)$; Lemma 1), so the joint intervention exists only as the quotient’s cluster arm, which must clamp X_2 into the box. Closed forms: $\mathbb{E}[Y \mid \text{do}(x_1)] = (2x_1 - 4)^2 + 0.04$, so $y^* = 0.04$ at $\text{do}(X_1=2)$ — driving X_2 ’s natural response to 4, outside the box; $\mathbb{E}[Y \mid \text{do}(x_2)] = (x_2 - 4)^2$, best over the box = 4; hence $V(\Pi) = 4$ and the price $V(\Pi) - y^* = 3.96$ (Prop. 2; premise (i) of Prop. 3 fails by construction). The declared hierarchy splits $C_1 = \{X_1, X_2\}$ into singletons; one split reaches the identity partition, whose MIS contains the optimal $\text{do}(X_1)$ arm.

Structural tests and oracle gates. Dataset-free structural tests assert, per condition, the exact fine MIS, the C-DAG signature (in)equalities, and the prior-tier flips claimed above, and end-to-end regressions assert QCBO’s trajectory invariance (arm choices and intervention values exactly; incumbents to 10^{-8}) under every protected condition and its divergence under A3. The data generators refuse to write datasets whose Monte-Carlo arm values, correlations, or policy-box geometry contradict the closed forms.

C Fixed-objective protocol: per-method details

The invariant is that every method optimizes the *same* objective, the true SCM’s noise-free `intervention_function`, and only the *structure each method reasons with* is perturbed. We realize this per method as follows.

QCBO (finest and coarse). The objective is read from the original graph’s SEM, while the structure is supplied separately through the assumed graph name (`CoarsenedGraph(..., assumed_graph_name=variant)`). Swapping the assumed graph changes the quotient, the exploration set, the identification formulas, and the prior, but never the SEM that generates outcomes. *Prior pipeline.* An identifiable arm’s prior mean instantiates the identified formula as a pipeline of GP-based plug-ins (backdoor $\rightarrow$ front-door $\rightarrow$ g -computation) on the bidirected-edge expansion of $\mathcal{G}^{\text{II}}$. A non-identifiable arm — and any identifiable query the sound-but-incomplete pipeline cannot instantiate — shares the common uninformative prior, the observational mean of Y . Because clusters are manipulable-only and interventions assign full clusters, the cluster-level formulas expand exactly into fine-grained regressions on observational data.

CBO (full-DAG identification). CBO is QCBO’s backend at the identity partition (Prop. 5): it runs the same complete identification algorithm, only on the full DAG rather than a quotient. The injection is the same assumed-graph swap as for QCBO (the perturbed structure reaches the exploration set and the identification formulas that build the prior), while outcomes are always drawn from the true SEM. Its trajectory is identical to QCBO’s at the finest partition on every graph.

Exploration-set disclosure. Both methods use the MIS of the assumed graph (Eq. 2), computed by the released code and verified against hand-derived Lee–Bareinboim ground truth per benchmark (for the minimal suite, the per-condition sets are also asserted by the structural tests of App. B). The CBO authors’ shipped code deviates from that definition in two places, which we correct for both methods in the family suite: on ToyGraph it lists the non-minimal arm $\{X, Z\}$ (mutilating Z cuts $X \rightarrow Z$, so $\text{do}(X, Z) \equiv \text{do}(Z)$ and the joint arm is redundant), and on CoralGraph it truncates the list at subsets of size three (an ad-hoc cap; the definition yields all 31 non-empty subsets, as every manipulable variable keeps a directed path to Y through never-intervened non-manipulables under any mutilation).

HQCBO. Phase 1 is the QCBO configuration run by the same code path with the same seed (the trigger is a function of the coarse trajectory alone), so its pre-trigger prefix is byte-identical to the QCBO run — asserted per seed. The refined phase re-enters the same backend with the split partition, the unioned exploration set, and the carried per-arm data (§3.2); the vendored loop forces an observation step at the start of a phase, a small bias *against* refinement that we keep.

Objective-identity verification. For all methods the decoupling holds by construction: outcomes are always evaluated on the true graph’s SEM, and the perturbation enters only through the separately-supplied assumed-graph variant (including assumed latents, as in A3 and B1). The released data generators additionally refuse to build any dataset whose Monte-Carlo arm values contradict the closed-form oracles (App. B): feeding a method a wrong graph changes only its causal reasoning, never the problem it is solving.

D Dataset graphs and their quotients

For every dataset in the paper, the figures below show the fine DAG with its coarse partition and the quotient C-DAG the method actually consumes — regenerated from the released structure registry by `scripts/emit_dataset_tikz.py`, so the pictures cannot drift from the code. On PARALLEL PARENT, note how the intra-cluster latent $X_1 \leftrightarrow X_2$ disappears in the quotient while $C_1 \rightarrow Y$ survives any single parent-edge deletion — the redundancy behind Experiment A; the FRONTDOOR and ToyGraph pairs both quotient to the bow of Fig. 3 — the exposure trade of Prop. 7.

E Implementation and reproducibility

All methods share one surrogate/acquisition backend (the vendored CBO loop), so comparisons differ only in the assumed structure. **Surrogate.** Each exploration-set arm carries a GPY `GPRegression` model with an RBF kernel (signal variance and lengthscale initialized to 1, ARD off) and observation noise fixed near zero (10^{-10} ,



Figure 7: **ParallelParent (MinimalBench, Exp. A)**. Fine DAG with coarse partition (left) and quotient C-DAG (right), in the convention of Fig. 1: manipulable variables orange, target blue, dash-dotted latent confounding; dashed boxes are the manipulable clusters. The intra-cluster $X_1 \leftrightarrow X_2$ disappears in the quotient; deleting $X_1 \rightarrow Y$ leaves $C_1 \rightarrow Y$ alive through $X_2 \rightarrow Y$ (quotient-redundant).



Figure 8: **FrontDoor (MinimalBench, Exp. B)**. Fine DAG with coarse partition (left) and quotient C-DAG (right), in the convention of Fig. 1: manipulable variables orange, target blue, dash-dotted latent confounding; dashed boxes are the manipulable clusters. The quotient is a bow ($C_1 \rightarrow Y$, $C_1 \leftrightarrow Y$): the cluster arm runs on the uninformative prior tier in every condition, so the assumed intra-cluster $X_1 \leftrightarrow M$ cannot change it.

matching the deterministic mechanisms). The causal prior supplies the do-calculus estimate as the GP *mean function*, so Δ -metrics isolate the effect of a wrong prior/exploration set from surrogate noise. **Acquisition.** Causal expected improvement [Garnett, 2023], maximized over the box-constrained intervention domain by gradient-based L-BFGS. **Protocol (minimal suite).** Unit per-variable intervention costs, 3 initial interventional points per arm, $T=50$ trials (60 on MEDIATEDCHAIN), 30 seeds, $n_{\text{obs}}=100$ from one shared dataset per SCM; the family suite follows each base paper’s own protocol (§5.4). Every run additionally logs the selected arm and intervention values per trial, so invariance is checked on *decisions*, not only on incumbent values. **Invariance tolerance.** Arm-choice and intervention-value sequences are compared exactly; incumbent trajectories at 10^{-8} (GP-solver floating point). On every protected condition of Table 2 the sequences are identical and $\sup |\Delta| = 0$; the quotient-visible control moves trajectories on every seed. **Artifacts.** The MinimalBench structural module (`ccbo/minibench.py`), its data generators with built-in oracle gates, the dataset-free structural tests and end-to-end invariance regressions, the suite runner (`scripts/run_minimal_suite.py`), the native two-phase HQCBO (`ccbo/minimal_suite.py`), and every table/figure emitter are released, so all reported numbers regenerate from the released SCMs and seeds.

F Full results

Table 5 positions QCBO in the method landscape (§6). Tables 4 and 6 give the per-environment detail behind §5.4’s model-based rows: finals and wall-clock on MCBO’s own tasks, and the within-family invariance counts under the intra-cluster perturbation of the model’s graph view. Per-seed CSVs for every minimal-suite unit (trial, incumbent, cumulative cost, selected arm, and intervention values) regenerate from the released runners (App. E).

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Figure 9: **MediatedChain (MinimalBench, Exp. C)**. Fine DAG with coarse partition (left) and quotient C-DAG (right), in the convention of Fig. 1: manipulable variables orange, target blue, dash-dotted latent confounding; dashed boxes are the manipulable clusters. The policy box restricts $D(X_2)$ to $[-2, 2]$; the fine optimum $\text{do}(X_1=2)$ drives X_2 to 4, outside the box.



Figure 10: **ToyGraph (CBO family)**. Fine DAG with coarse partition (left) and quotient C-DAG (right), in the convention of Fig. 1: manipulable variables orange, target blue, dash-dotted latent confounding; dashed boxes are the manipulable clusters. The quotient is a bow ($C_1 \rightarrow Y$, $C_1 \leftrightarrow Y$): $\text{do}(C_1)$ is not identifiable from the C-DAG, so the arm runs on the uninformative prior tier.

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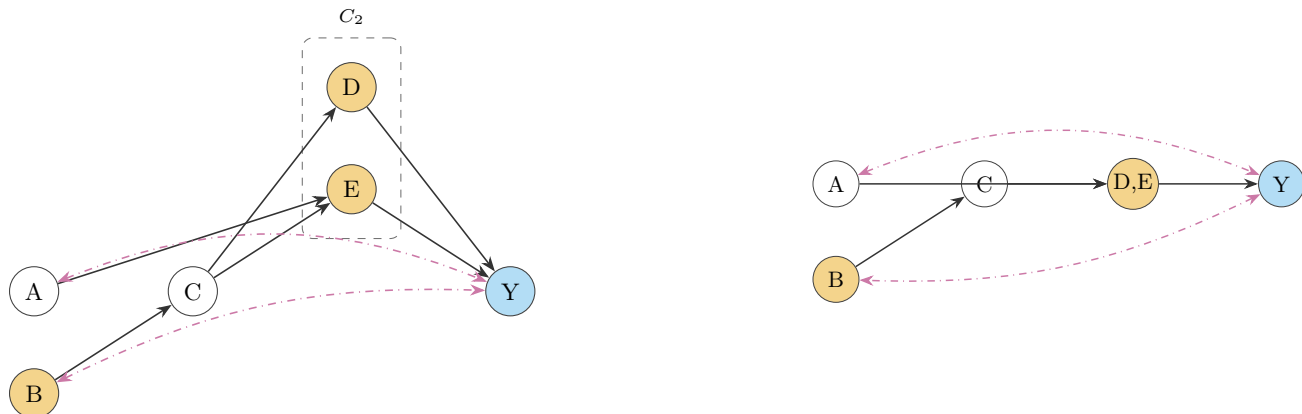


Figure 11: **CompleteGraph (CBO family)**. Fine DAG with coarse partition (left) and quotient C-DAG (right), in the convention of Fig. 1: manipulable variables orange, target blue, dash-dotted latent confounding; dashed boxes are the manipulable clusters.

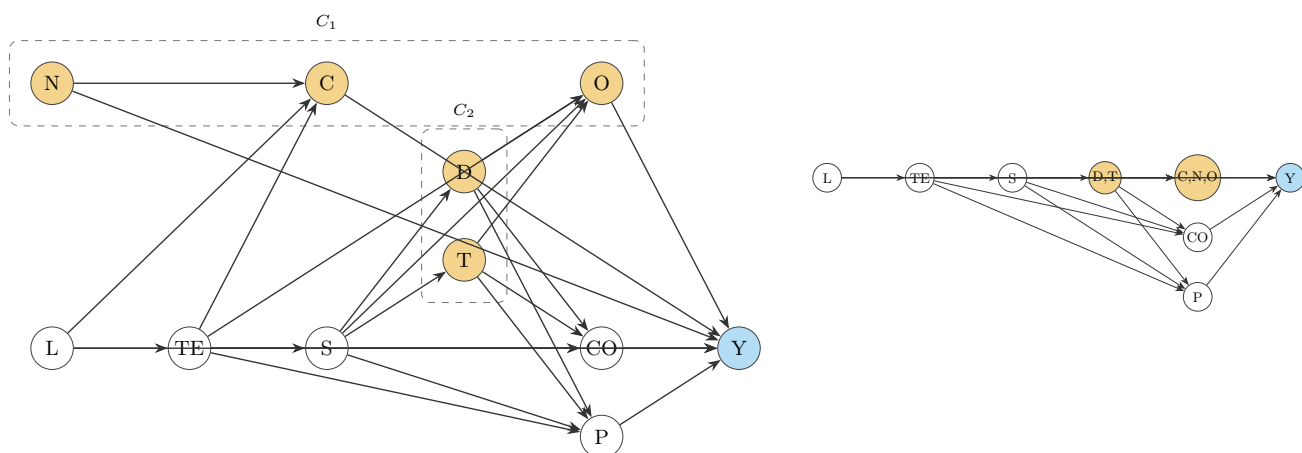


Figure 12: **SimplifiedCoralGraph (CBO family)**. Fine DAG with coarse partition (left) and quotient C-DAG (right), in the convention of Fig. 1: manipulable variables orange, target blue, dash-dotted latent confounding; dashed boxes are the manipulable clusters.

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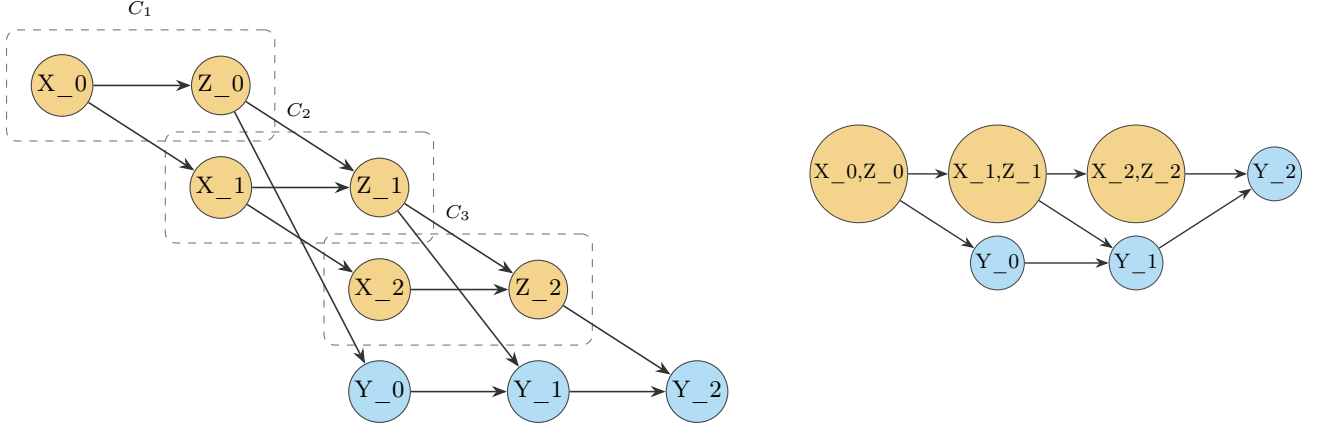


Figure 13: **DCBO stat / nonstat (QDCBO family)**. Fine DAG with coarse partition (left) and quotient C-DAG (right), in the convention of Fig. 1: manipulable variables orange, target blue, dash-dotted latent confounding; dashed boxes are the manipulable clusters. Slices $t = 0, 1, 2$; the nonstat setup shares this topology with a change point in the SEM.

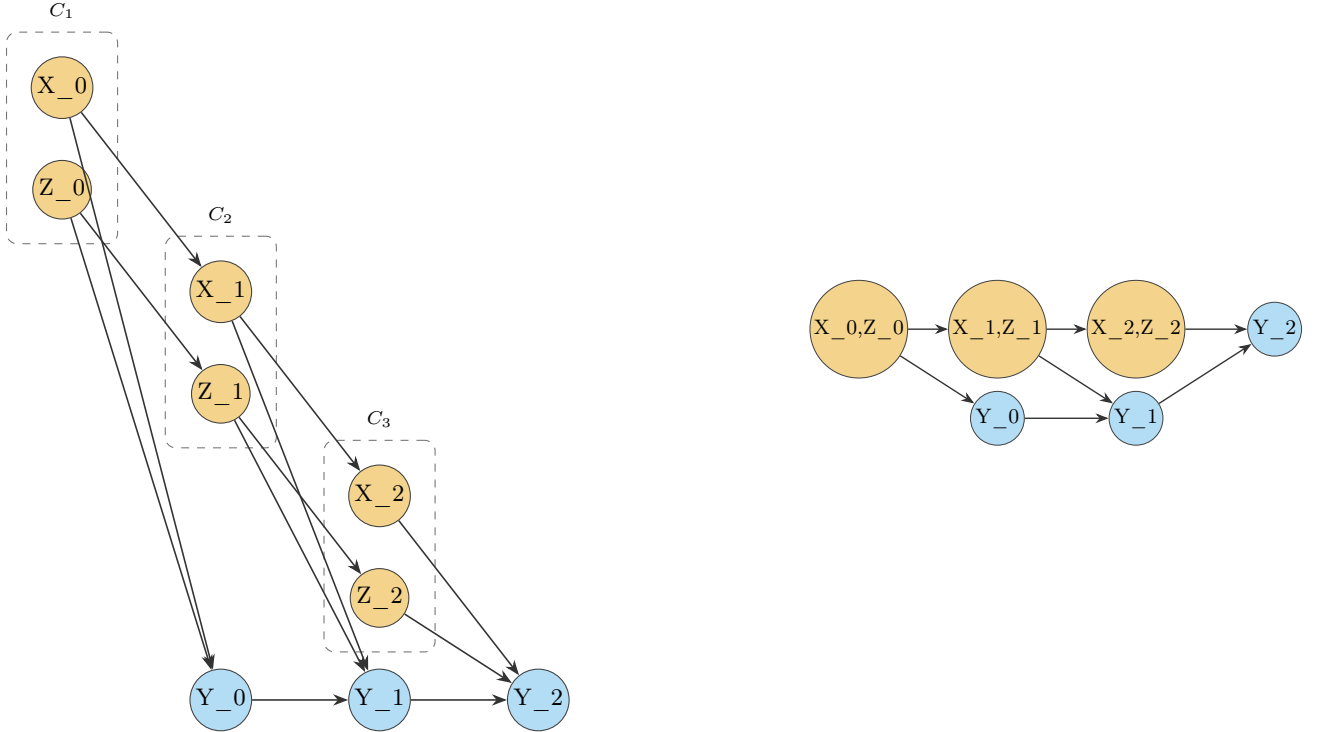


Figure 14: **DCBO ind (QDCBO family)**. Fine DAG with coarse partition (left) and quotient C-DAG (right), in the convention of Fig. 1: manipulable variables orange, target blue, dash-dotted latent confounding; dashed boxes are the manipulable clusters.

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Figure 15: **ToyGraph-M (QMCBO family)**. Fine DAG with coarse partition (left) and quotient C-DAG (right), in the convention of Fig. 1: manipulable variables orange, target blue, dash-dotted latent confounding; dashed boxes are the manipulable clusters.

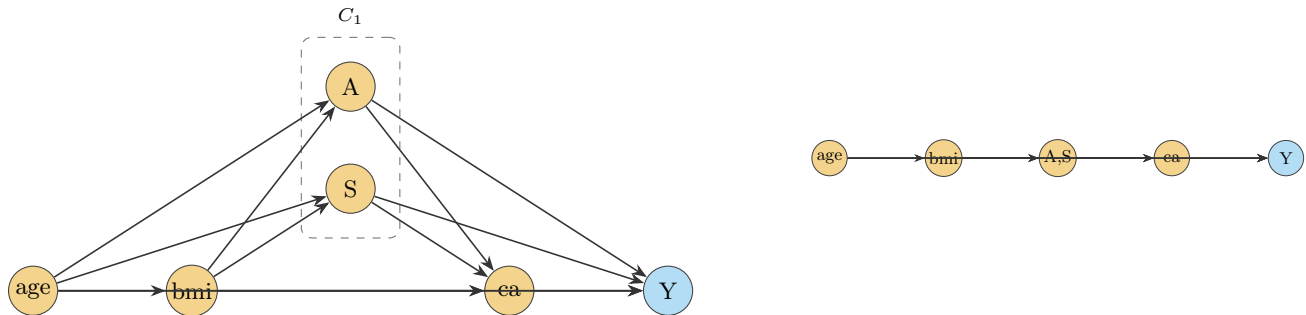


Figure 16: **PSAGraph (QMCBO family)**. Fine DAG with coarse partition (left) and quotient C-DAG (right), in the convention of Fig. 1: manipulable variables orange, target blue, dash-dotted latent confounding; dashed boxes are the manipulable clusters.

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Algorithm 1 QCBO: Quotient Causal Bayesian Optimization

Require: C-DAG $\mathcal{G}^\Pi$ over the clusters of Π (or a fine hypothesis $\mathcal{G}$ and Π); target Y ; number of trials T ; observational data $\mathcal{D}^O$; optional hierarchy H (§3.2).

- 1: If given $\mathcal{G}$: $\mathcal{G}^* \leftarrow$ latent projection of $\mathcal{G}$; $\mathcal{G}^\Pi \leftarrow$ quotient of $\mathcal{G}^*$ under Π . Check acyclicity of $\mathcal{G}^\Pi$ (Assumption 1).
- 2: Build $\mathbf{ES}^\Pi \leftarrow \text{MIS}(\mathcal{G}^\Pi)$ via Eq. (3).
- 3: For each $\mathbf{X}_s \in \mathbf{ES}^\Pi$: run complete ID on $P(Y \mid \text{do}(\mathbf{X}_s))$ and fit the prior mean m_s via the two-tier rule, Eq. (4), on $\mathcal{D}^O$.
- 4: Collect initial interventional data $\mathcal{D}_0^I$.
- 5: **for** $t = 1, \dots, T$ **do**
- 6: Update GP posteriors $(m_{s,t}, k_{s,t})$ for all $\mathbf{X}_s \in \mathbf{ES}^\Pi$.
- 7: $(\mathbf{X}_{s_t}, \mathbf{x}_t) \leftarrow \arg \max_{s, \mathbf{x}} \text{EI}_t^s(\mathbf{x}) / \text{Co}(\mathbf{X}_s, \mathbf{x})$.
- 8: Intervene, observe y_t ; update $\mathcal{D}_t^I$.
- 9: **if** H declared **and** incumbent plateaued: split the incumbent’s clusters via H , re-check validity, rebuild $\mathbf{ES}^\Pi \cup$ past arms, carry $\mathcal{D}_t^I$ by arm, init new arms (§3.2).
- 10: **end for**
- 11: **return** $(\mathbf{X}_{s_{t^*}}, \mathbf{x}_{t^*})$ where $t^* = \arg \min_t y_t$.

Table 4: QMCBO vs MCBO (its own harness and protocol; 5 seeds, $T=100$). Final Y (all tasks maximized) and wall-clock per unit.

Env	Method	Final Y	s/unit
ToyGraph ($y^*=2.172$)	MCBO	1.143 ± 0.271	1410
	QMCBO	2.150 ± 0.013	1570
PSAGraph ($y^* = -4.13$)	MCBO	-5.153 ± 0.000	6830
	QMCBO	-5.153 ± 0.000	2266

Table 5: Positioning of QCBO against the nearest methods.

Method	What is learned	Structural prior	Search space
CBO [Aglietti et al., 2020]	nothing (oracle DAG)	full DAG	MIS
GACBO [Mukherjee et al., 2024]	posterior over DAGs	DAG family	union of MIS
Durand et al. [Durand et al., 2025]	$\text{Pa}(Y) \subseteq \mathbf{X}$	Y leaf	$\mathbf{X}$
Park et al. [Park et al.]	$\text{PAG} \rightarrow \text{MAG}$	MEC	partial $\mathbf{X}$
SCB [Lee and Bareinboim, 2018]	nothing (oracle DAG)	full DAG	POMIS
QCBO (ours)	nothing	partition + C-DAG	identifiable cluster unions

Table 6: Invariance under an intra-cluster misspecification of the model’s graph view (same seeds, deterministic runs): trajectories identical to the correct-graph run, and mean $|\Delta_{\text{final}}|$. QMCBO cannot move (Theorem 1). MCBO does.

Env	Method	identical	$ \Delta_{\text{final}} $
ToyGraph	MCBO	2/5	0.299
	QMCBO	5/5	0.000
PSAGraph	MCBO	0/5	0.000
	QMCBO	5/5	0.000